# Magic Square Sudoku
# (The Golden Lo Shu Square)

magic@logicallayout.net
https://buymeacoffee.com/fuzzybit

September 13, 2026

**Abstract**

Sudoku with the use of a 3-by-3 Magic Square exhibits symmetric properties which can be utilized to solve the puzzle with 16 clues.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |
| | 1 | | | 2 | | | 3 | |
| | | | | | | | | |
| | | | 2 | 7 | 6 | | | |
| | 4 | | 9 | | 1 | | 6 | |
| | | | 4 | 3 | 8 | | | |
| | | | | | | | | |
| | 7 | | | 8 | | | 9 | |
| | | | | | | | | |

Figure 1: Sudoku with multiple solutions.

## 0.1 3-by-3 Grids

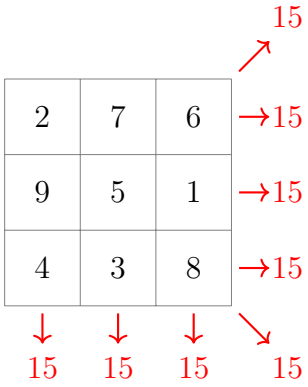



Figure 2: 3-by-3 Magic Square, grid (1,1) Sudoku

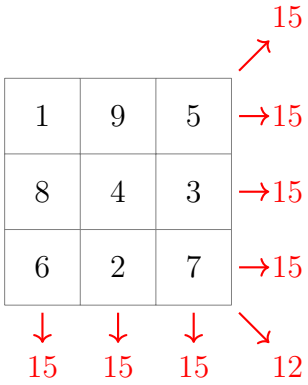



Figure 3: 3-by-3 Semi-Magic Square, grid (1,2) Sudoku

| | | |
|---|---|---|
| 3 | 8 | 4 |
| 7 | 6 | 2 |
| 5 | 1 | 9 |

Figure 4: 3-by-3 Semi-Magic Square, grid (1,3) Sudoku

| | | |
|---|---|---|
| 8 | 4 | 3 |
| 6 | 2 | 7 |
| 1 | 9 | 5 |

Figure 5: 3-by-3 Semi-Magic Square, grid (2,1) Sudoku

| | | |
|---|---|---|
| 7 | 6 | 2 |
| 5 | 1 | 9 |
| 3 | 8 | 4 |

Figure 6: 3-by-3 Semi-Magic Square, grid (2,2) Sudoku

| | | |
|---|---|---|
| 6 | 2 | 7 |
| 1 | 9 | 5 |
| 8 | 4 | 3 |

Figure 7: 3-by-3 Magic Square, grid (3,3) Sudoku

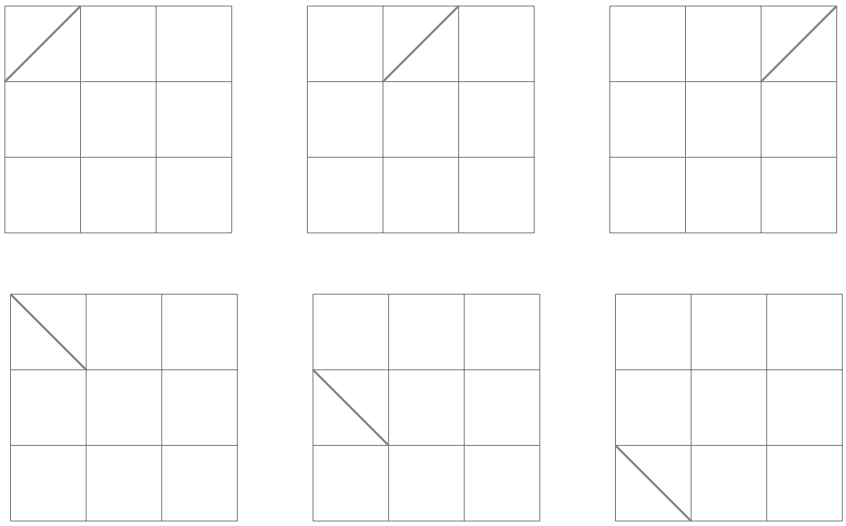
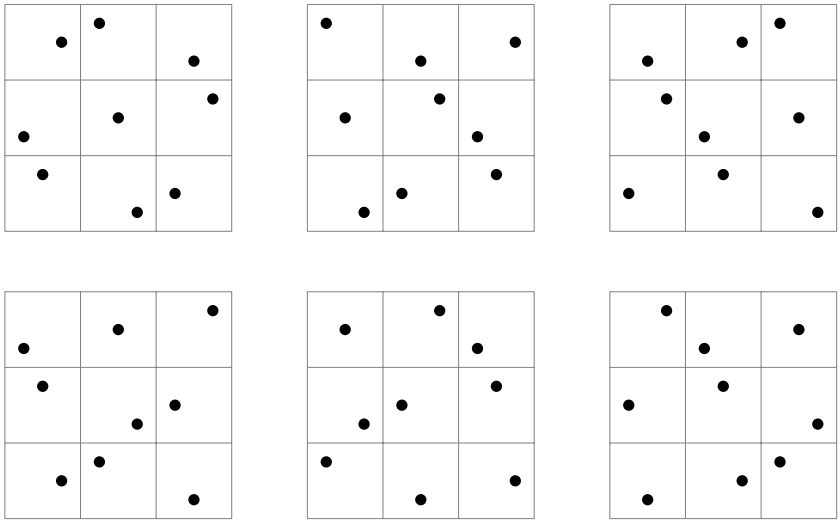


Figure 8: 3-by-3 Graphs of Diagonal Sums of 15



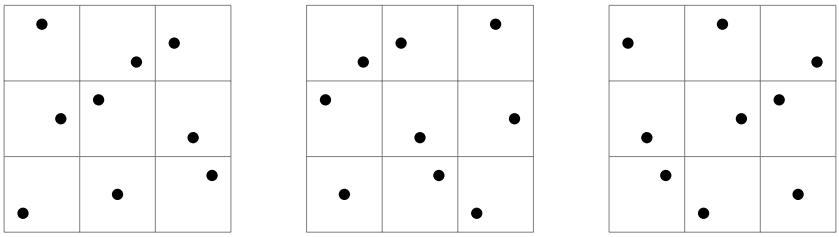


Figure 9: Singletons 1 to 9 in Sudoku Magic Square

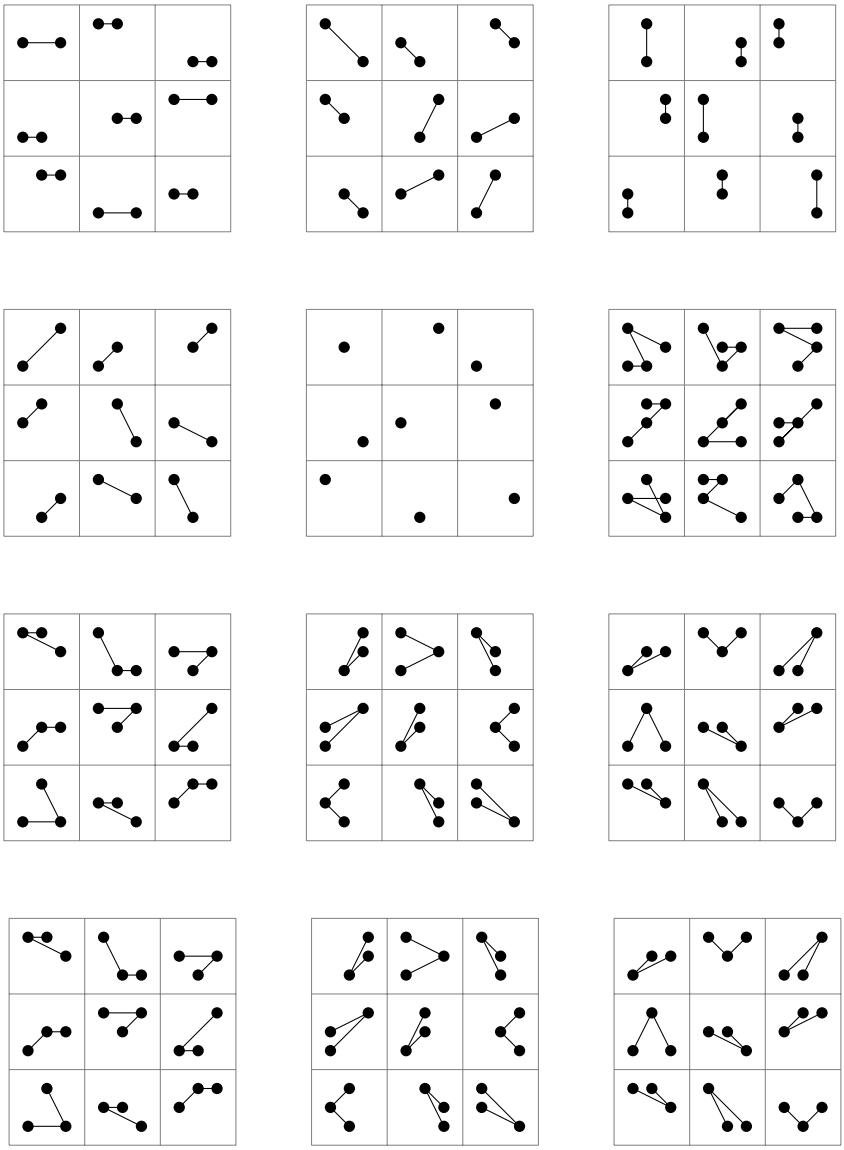


Figure 10: Singletons 1 to 9 in Sudoku Magic Square

# References

Andrews, W. S. (1908). *Magic Squares and Cubes*. With contributions by Paul Carus, Lorraine Screven Frierson, and Charles Albert Browne. Chicago: Open Court Publishing Company. URL: https://archive.org/details/magicsquarescube00andrrich.

McGuire, Gary, Bastian Tugemann, and Gilles Civario (2013). "There is no 16-Clue Sudoku: Solving the Sudoku Minimum Number of Clues Problem via Hitting Set Enumeration". In: *arXiv preprint arXiv:1201.0749*. URL: https://arxiv.org/abs/1201.0749.

Rosenhouse, Jason and Laura Taalman (2011). *Taking Sudoku Seriously: The Math Behind the World's Most Popular Pencil Puzzle*. Oxford; New York: Oxford University Press. ISBN: 9780199756568. URL: https://archive.org/details/takingsudokuseri0000rose/mode/2up.